

Introduction to Parallel Computing

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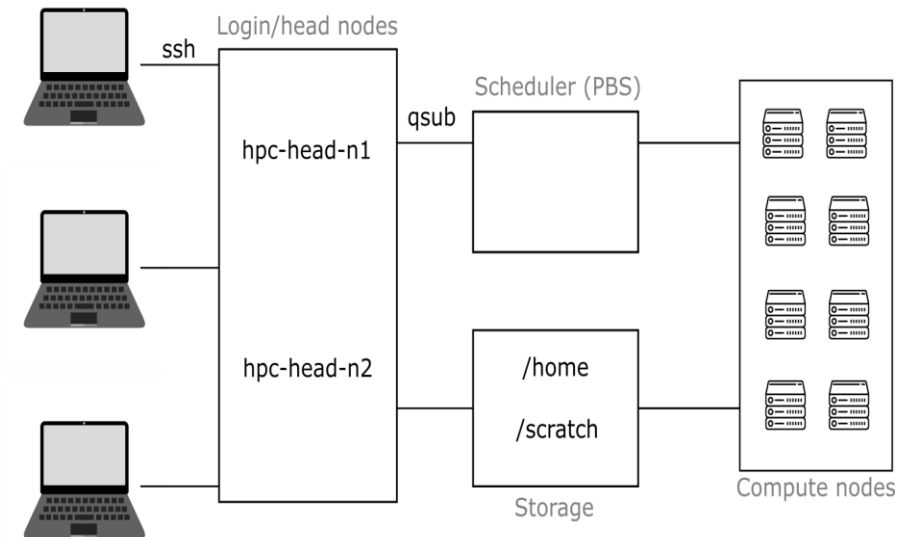
Lab 1
Introduction to Cluster

Overview

- **HPC Architecture:** Separation of Head (Login) and Compute nodes.
- **Access & Storage:** VPN/SSH login, file transfer (SCP), and home quota limits.
- **Linux & Software:** Essential terminal commands and Environment Modules.
- **Job Scheduling:** Queue management and job submission via PBS

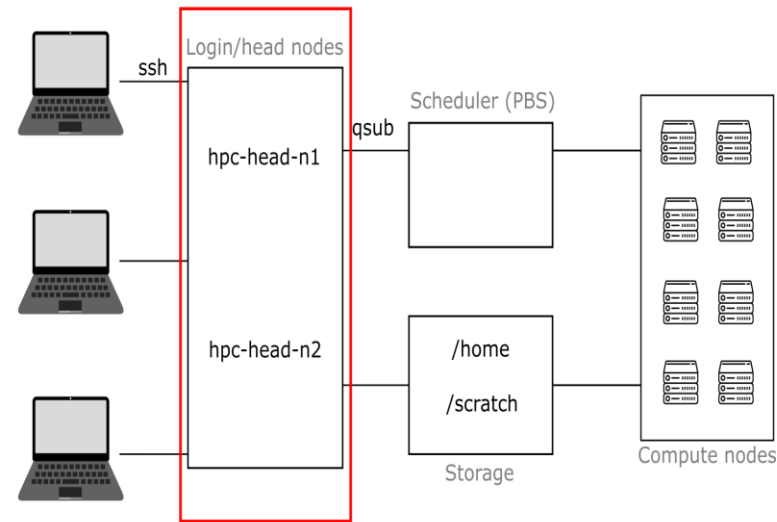
Cluster

- A collection of interconnected computers (nodes) that work together as a single system. These interconnected computers are endowed with software to coordinate programs on those computers and they can work together to perform computationally intensive tasks



HPC UniTn cluster architecture

Cluster Components and Setup

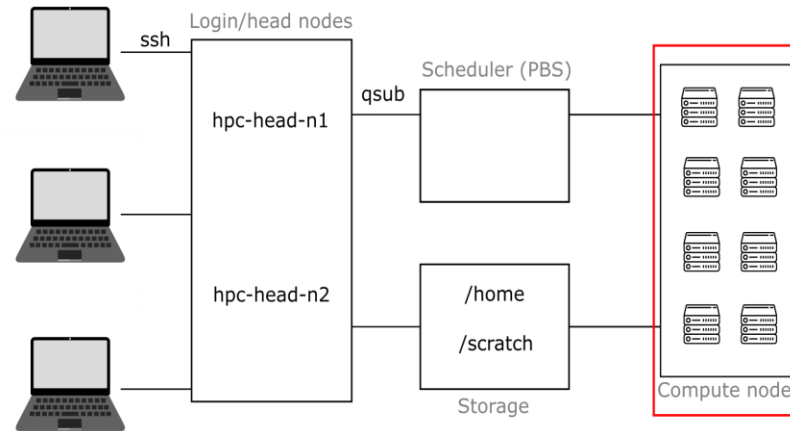


HPC UniTn cluster architecture

Head Nodes: Nodes where users log in to submit or manage their jobs

- Log in ✓
- Job submission, monitoring and management ✓
- Edit files and analyze your output ✓
- Compilation ?
- Running simulations ✗

Cluster Components and Setup

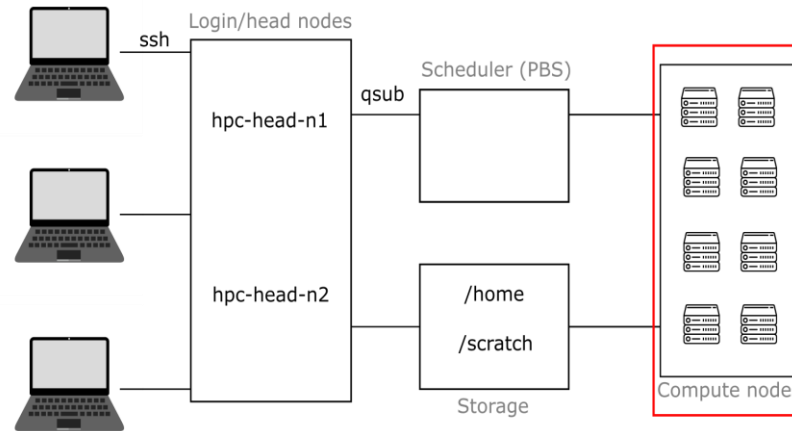


HPC UniTn cluster architecture

Compute Nodes: Nodes that perform the actual computation in parallel. Each node has its own processor(s) and memory and multiple compute nodes work together to solve parts of a larger problem

- Compilation ✓
- Running simulations ✓
- Run an interactive session ✓

Cluster Networking



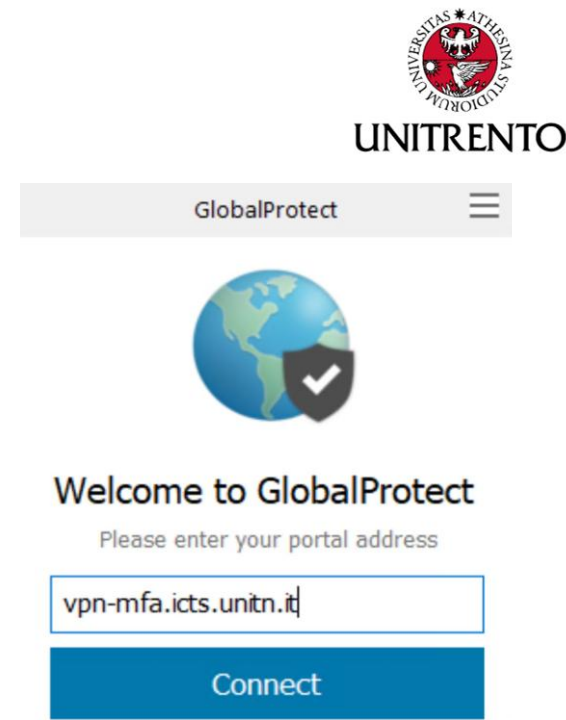
HPC UniTn cluster architecture

Interconnection Networks:

- Enables communication between nodes, critical for coordination and data sharing in parallel tasks.
 - Ethernet: Widely used, affordable, but with higher latency and lower bandwidth compared to specialized networks
 - InfiniBand: switch-based serial I/O interconnect with high-speed, low-latency networking technology
 - Omni-path: switch-based interconnect with low communication latency, low power consumption and a high throughput

Access and login

- Access to the login nodes is done using the SSH protocol.
- Use [GlobalProtect \(VPN\)](#) to ensure secure connection
- Portal: vpn-mfa.icts.unitn.it



To access the cluster:

- **MAC or Linux:** Open the Terminal (or Console) and use the SSH command [ssh username@hpc3-login.unitn.it](#)
- **Windows:** [PuTTY](#) and [Mobaxterm](#) are some of the most popular SSH clients for Windows.
- Once the SSH client has been installed, you can enter the server name you wish to connect to ([username@hpc3-login.unitn.it](#)) and enter your password to connect to the cluster.

Hardware description of UniTn cluster

Component	Description
Architecture	HPC cluster managed using the PBS job scheduling system
CPU	15,380 CPU cores
GPU	16 GPUs
RAM	143.15 TB total RAM
Ethernet	General network communication, cluster management, login access and data transfer
InfiniBand	High-speed, low-latency communication between compute nodes, particularly for MPI applications
Omni-Path	High-performance, low-latency interconnect for HPC/MPI communication; available on some nodes

Storage

- Each user has a home directory backed up on a performing storage.
- Maximum quota for the home is 200GB.
- Users who occupy more than 5Gb in their homes will be informed via email monthly.
- **Remove processing data** while keeping only execution codes in the home directory

Transfer files – SCP (Secure Copy Protocol)

- To copy a file from a local directory to the cluster

```
$scp /localpath/File Name username@hpc3-login.unitn.it:/remotepath/File Name
```

- To copy a directory, it is necessary to use the flag -r for recursion

```
$scp -r /localpath/DirName username@hpc3-login.unitn.it:/remotepath/DirName
```

- To transfer a file from the cluster to a local folder

```
$scp username@hpc3-login.unitn.it:/remotepath/File Name /localpath/ File Name
```

- To copy a directory

```
$scp -r username@hpc3-login.unitn.it:/remotepath/DirName /localpath/DirName
```

Basic Linux Commands

- **Pwd**: displays the absolute path to the directory which you are in
- **Ls**: you can use ls to display what files are in the directory you are in
 - ls -a: display all the hidden files
 - ls -l: display the file permissions
 - ls -lah: include the file sizes
- **cd**: the command cd is used to go to a directory. If you type cd and press enter, it takes you to your home directory. To navigate up one directory level, you can type cd .. (the two dots means up one level)
- **mkdir & rmdir**: mkdir is used to create a folder and rmdir to delete a directory (only for empty directories)
- **rm**: deletes files and directories. If you use rm -r, you delete both the folder and the files contained in this folder
- **touch filename**: If the filename that it is given does not exist yet, it creates a file of that name as a new and empty file.

Basic Linux Commands

- **cp:** copies files. It takes two arguments: the location of the file to be copied and the place where the file will be copy
- **mv:** moves files. It takes the same two arguments than cp
- **man & -help:** both are used to know more about a command and how to use it. man shows the manual pages of the command and typing the command name and the argument help you can see how to use the command
- **top:** shows what is happening in real time on the system (you can sort the output by memory usage (M), by CPU usage (P), by CPU time (T))
- **du -h:** display the disk usage (in Gb) of a file
- Please, use **du -hs** on your HOME to check your total disk usage and remove the unnecessary files

Computing nodes

- Computing nodes are those accessed through a batch system and where jobs are executed.

Resource	Public Queues	Private Queues
CPU	shortCPUQ	cibiocmCPUQ
	commonCPUQ	cibioCPUQ
	longCPUQ	dsCPUQ
		montresorCPUQ
		dicamCPUQ
		variamolsCPUQ
		mathCPUQ
		mathNUMQ
		variamolsGPUQ
GPU	shortGPUQ	mathGPUQ
	commonGPUQ	particleGPUQ
		cibioGPUQ
		bcbGPUQ

Computing nodes

- Public queues (available for all users) include all cluster CPU nodes and those bought by the University for shared use.

Queue	Max concurrent jobs/user in queue	Max concurrent jobs/user running	Max wall time	Access
shortCPUQ	30	30	6 h	Free
commonCPUQ	–	30	100 h	Free
longCPUQ	10	10	—	Free

- Private queues contain CPU nodes owned by specific department.

PBS (Portable Batch System)

- PBS is a job scheduler and resource manager used in High-Performance Computing.
- It manages jobs and assigns computing resources such as CPU, memory, and GPU.
- Basic workflow:
User → Submit Job → Queue → Compute Node → Job Runs

Command	Purpose
<code>qsub job.pbs</code>	Submit a PBS job
<code>qstat</code>	Show current jobs
<code>qstat -u \$USER</code>	Show your jobs
<code>qstat -Q</code>	Show available queues
<code>pbsnodes -aS</code>	Show node status (summary)
<code>qdel JOB_ID</code>	Cancel a job
<code>qsub -l</code>	Start an interactive job

Software on HPC

- Software is loaded using modules
 - To display the available program environment modules

```
[username@hpc-head-n1 ~]$ module avail
```

- To load some program environment

```
[username@hpc-head-n1 ~]$ module load module Name
```

- To display the loaded modules

```
[username@hpc-head-n1 ~]$ module list
```

- To remove environment variables of a loaded module

```
[username@hpc-head-n1 ~]$ module unload module Name
```

- To purge all loaded modules

```
[username@hpc-head-n1 ~]$ module purge
```

Job processing

- To start an interactive session, `qsub -l` is used.

```
[ username@hpc-head-n1 ~]$ qsub -l -q queueName [more options]
```

- To exit of the interactive session:

```
[ username@hpc-c04-node03 ~]$ exit
```

- Submit the job to the queue using a batch script

```
[ username@hpc-head-n1 ~]$ qsub [options] scriptName.pbs
```

- The submission script must contain
 - the application to be executed
 - queue name
 - resources required to execute the job (number of nodes, number of CPUs/threads, memory, etc.)

Managing jobs

- Job status: To view all the jobs of a user, use the command `qstat -u`

```
[ username@hpc-head-n1 ~] $ qstat -u userName
```

- Status of the job: E (exiting after having run), H (held), Q (queued), R (running), W (waiting)
- To view the detailed status of a single job use `qstat -f`

```
[ username@hpc-head-n1 ~] $ qstat -f $PBS_JOBID
```

- Deleting job To cancel a job use the command `qdel`

```
[ username@hpc-head-n1 ~] $ qdel $PBS_JOBID
```

Bash job submission: Hello (in C)

```
#!/ bin / bash
#PBS -N hello_c      # Job name
#PBS -o ./hello_c.out # Output files
#PBS -e ./hello_c.err # Error files
#PBS -q shortCPUQ    # Queue name
#PBS -l walltime=0:02:00 # Set the maximum wall time
#PBS -l select=1:ncpus=1
#PBS -l mem=1mb # Set the memory # Load modules
module load GCC/13.3.0
# Select the working directory
cd / home / username / testFolder
# Compile and run ( sequential code )
gcc hello.c -o hello
./ hello
```

Measuring Execution Time

- **gettimeofday()** is used to measure the wall-clock execution time of a piece of code.
- It stores time in a **struct timeval**:
 - **tv_sec** → seconds
 - **tv_usec** → microseconds
- **Steps:**
 - Get the **start time** before the code.
 - Run the code/function.
 - Get the **end time** after the code.
 - Subtract the two times.

```
struct timeval start, end;

gettimeofday(&start, NULL);

array_addition(a, b, c, n); // code to measure

gettimeofday(&end, NULL);

double elapsed =
    (end.tv_sec - start.tv_sec) +
    (end.tv_usec - start.tv_usec) / 1000000.0;

printf("Time = %.6f seconds\n", elapsed);
```

Exercise

- Create a HelloWorld program in C, Compile and execute on the cluster
- Create a array addition program in C, Compile and run on the cluster
 - Create a bash script for submitting the jobs and execute them
 - Execute them on an interactive session

Key Takeaway

- **Head vs. Compute Nodes:** Manage jobs on Head nodes; run all computations on Compute nodes.
- **Secure Access:** Connect via VPN + SSH (`hpc3-login.unitn.it`).
- **Storage Limit:** Keep `/home` under 200GB by removing temporary output files.
- **PBS & Modules:** Use `qsub` for jobs (`qsub -I` for interactive) and `module load` to enable required tools.